
Project Proposal: Type Rhythm Game



Presented By

Maliha Tabassum
Id : 20231078010
Section : B
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Presented To

Abu Naim Khan
Lecturer
North Western University, Khulna

1. Introduction

The Type Rhythm Game is a browser-based mini-game where words fall from the top of the screen, and the player must type them correctly before they hit the bottom. This game aims to enhance users' typing speed and reflexes by making the experience interactive and engaging. It uses core web technologies—HTML, CSS, and JavaScript—to simulate the falling word animation, handle real-time input detection, and maintain game logic such as scoring and level changes. The concept is simple yet challenging, designed to test and improve players' quick thinking and keyboard coordination with varying levels of difficulty.

2. Literature Review

- Why: Typing games have been widely used for learning and improving typing skills. However, most games focus only on speed and lack rhythm-based engagement.
- How: This project introduces rhythm mechanics through falling speed and background changes.
- Related Topics: Gamification in learning, typing tutor applications, rhythm-based cognitive training.
- Gaps: Lack of typing games that blend rhythm, speed, and visual engagement for casual learners.

3. Methodology

- How:
 1. HTML: for building the structure of the game (scoreboard, input box, difficulty select, word container).
 2. CSS: for layout, animation of falling words, and responsive design.
 3. JavaScript: for word spawning logic, speed control, input checking, score tracking, and game over handling.
- Game Flow:
 1. User selects a difficulty.
 2. Words fall periodically.
 3. Typing correct word removes it and gives points.
 4. Missed words deduct points.
 5. Score below zero ends the game.
- Tools: Browser environment only. No frameworks or back-end used.

4. Preliminary Data

- Early test runs showed:
 - Users enjoy increasing challenge with faster word drop.
 - Players type faster after 3–5 rounds.
 - Engagement improves with visual effects and cleaner design.

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- Important Categories:
 - Difficulty level
 - Score behavior
 - Word frequency

5. Statement of Limitations

- No real-time multiplayer or online leaderboard.
- May not suit visually impaired users.
- Sound and accessibility features are minimal.

6. Conclusion

The Type Rhythm Game is a simple yet unique typing game that incorporates rhythm-based elements to create an enjoyable and productive experience. Developed using fundamental frontend technologies, the game effectively demonstrates how JavaScript can be used to build interactive browser applications. It motivates users to practice their typing while providing immediate feedback and growing difficulty. Overall, this mini-project reflects the creative application of web development skills in building a practical and fun educational tool.